

# ImpactSphere

AI-Powered CSR Need-to-Impact Allocation & Optimization Platform

“Don’t just fund good projects. Fund the right projects, in the right places, for the people who need them most.”

## **Comprehensive Project Report**

Hackathon / Prototype Blueprint

# 1. Executive Summary

ImpactSphere is an AI-powered CSR decision-intelligence platform designed to help Indian companies make more meaningful, transparent, data-driven and compliant CSR funding decisions.

The central problem is not simply that companies need to spend CSR money. The deeper problem is: **given a limited CSR budget and many project proposals, how can a company identify where its money can create the greatest meaningful social impact, especially for communities with greater unmet needs?**

## 2. Problem Statement

CSR teams may need to evaluate many project proposals while considering cost, beneficiaries, location, social need, implementation feasibility, Schedule VII alignment and other CSR requirements. Manual evaluation can become spreadsheet-heavy, slow and subjective.

Example: a company has ₹5 Crore available for CSR and receives 20 proposals. A project serving more beneficiaries is not automatically the best use of funds if another project addresses a substantially larger unmet need.

## 3. Core Objective

Help CSR committees allocate limited funds toward projects that are **compliant, impactful, cost-effective, aligned with community needs, geographically balanced and capable of producing measurable social outcomes.**

ImpactSphere is therefore not merely a CSR management dashboard or an AI proposal-ranking tool. It is **CSR Need-to-Impact Decision Intelligence.**

## 4. Central Innovation

ImpactSphere connects four dimensions that are often handled separately: **where the need is, what projects are available, which projects are eligible, and where money should go.**

NEED  
+  
PROJECT  
+  
COMPLIANCE  
+  
IMPACT  
+  
BUDGET  
↓  
BEST PORTFOLIO

## 5. Need-to-Impact Concept

Traditional evaluation asks: “Which projects are good?” ImpactSphere asks: “Which eligible projects can create the greatest additional impact for communities with the greatest unmet needs, given the available budget?”

WHERE?  
↓  
District / Region

WHAT NEED?  
↓  
Healthcare / Education / Water / Livelihood etc.

HOW SEVERE?  
↓  
Need Index

WHAT PROJECT?  
↓

Available CSR proposal

WHAT IMPACT?

↓

Expected outcomes

HOW MUCH?

↓

Project cost

WHAT CAN WE AFFORD?

↓

Optimization

WHAT ACTUALLY HAPPENED?

↓

Outcome measurement

## 6. Social Need Index

Beneficiary count alone is not enough. A project reaching 1,000 people in an extremely underserved area may deserve greater priority than a project reaching 5,000 people in an already well-served area.

The Need Index represents the relative severity of unmet development needs in a region for a particular intervention area. It should be based on transparent indicators rather than invented by an AI model.

Healthcare Need  
Education Need  
Water/Sanitation Need  
Livelihood Need  
Infrastructure Need  
Environmental Need  
↓  
SOCIAL NEED INDEX

| Region     | Healthcare | Education | Water | Overall Need |
|------------|------------|-----------|-------|--------------|
| District A | 91         | 84        | 88    | 88           |
| District B | 62         | 70        | 54    | 62           |
| District C | 38         | 45        | 41    | 41           |

## 7. Real-World Data Foundation

For the hackathon, curated or synthetic data can be used. For a real implementation, public development indicators can feed the Need & Gap Engine.

NITI Aayog's Aspirational Districts Programme is a relevant foundation: it focuses on 112 under-developed districts and uses 49 Key Performance Indicators across Health & Nutrition, Education, Agriculture & Water Resources, Financial Inclusion & Skill Development, and Infrastructure.

## 8. Unmet Need and Gap Detection

Need alone is not sufficient. ImpactSphere should estimate how much of that need is already being addressed and identify the remaining gap.

A region with slightly lower overall need but much lower existing coverage may have the larger actionable gap.

District A  
Healthcare Need = 90  
Existing coverage = 80  
Unmet Need = 10

District B  
 Healthcare Need = 85  
 Existing coverage = 30  
 Unmet Need = 55

Priority  
 =  
 Need Severity  
 ×  
 Unmet Need  
 ×  
 Expected Impact  
 ×  
 Cost Efficiency

## 9. Avoiding Duplication

The platform can detect situations where CSR interventions are heavily concentrated in one sector or region. This provides portfolio-level context and can highlight under-addressed areas such as water, sanitation, education or livelihoods.

## 10. AI Proposal Intelligence

The AI layer converts unstructured project proposals into structured information that downstream deterministic systems can use.

Typical extracted fields include project name, location, budget, beneficiaries, duration, target population, intervention type and expected outcomes.

PDF / Proposal  
 ↓  
 Text Extraction  
 ↓  
 LLM / NLP Layer  
 ↓  
 Structured Information  
 ↓  
 ML Classification  
 ↓  
 Confidence + Evidence  
 ↓  
 Compliance + Scoring

## 11. AI Classification and Model Training

The model-training work should focus on a narrow, measurable task rather than attempting to train a large language model from scratch.

**Dataset:** create proposal examples covering relevant Schedule VII/intervention categories; include title, description, objective, location, beneficiaries, budget and outcomes; split into train, validation and test sets.

**Baseline:** TF-IDF + Logistic Regression. **Improved approach:** Sentence Transformer embeddings + classifier. Evaluate accuracy, precision, recall and F1-score.

## 12. AI + LLM Hybrid Architecture

The robust architecture is hybrid. The LLM understands proposal language, while deterministic software performs statutory calculations and compliance logic.

| Layer | Primary Responsibility                                             |
|-------|--------------------------------------------------------------------|
| LLM   | Information extraction, summarization, expected-outcome extraction |

| Layer          | Primary Responsibility                                     |
|----------------|------------------------------------------------------------|
| ML model       | Classification, semantic similarity, confidence estimation |
| Rule engine    | CSR calculations and deterministic compliance logic        |
| Optimizer      | Budget allocation and portfolio constraints                |
| Human reviewer | Final judgement, overrides and approval                    |

## 13. CSR Compliance Engine

The compliance layer evaluates proposals against the applicable Indian CSR framework. The core legal framework is based on Section 135 of the Companies Act, 2013, Schedule VII and the Companies (CSR Policy) Rules, 2014, as amended.

Starting applicability inputs include: Net Worth ≥ ₹500 Cr OR Turnover ≥ ₹1,000 Cr OR Net Profit ≥ ₹5 Cr. The statutory calculation and legal checks should be implemented deterministically and reviewed for current amendments.

Schedule VII categories and rules should be stored as versioned data, not scattered throughout the frontend. Ambiguous cases should be marked for human review.

```
Net Worth ≥ ₹500 Cr
OR
Turnover ≥ ₹1,000 Cr
OR
Net Profit ≥ ₹5 Cr
```

## 14. Rule Database

Recommended schema: `csr_rules(rule_id, source, section, rule_name, rule_type, rule_value, description, effective_from, effective_to, version)`.

Versioning allows the system to update rules when the CSR framework changes and makes recommendations auditable.

## 15. Impact Scoring

An example configurable scoring model is: Need/Vulnerability 25%, Expected Social Impact 25%, Beneficiary Reach 15%, Cost Efficiency 15%, CSR Alignment 10%, Feasibility 5%, Sustainability 5%.

These weights are decision preferences, not statutory CSR requirements. The system should record the weights used for each optimization run.

| Factor                 | Weight |
|------------------------|--------|
| Need / Vulnerability   | 25%    |
| Expected Social Impact | 25%    |
| Beneficiary Reach      | 15%    |
| Cost Efficiency        | 15%    |
| CSR Alignment          | 10%    |
| Feasibility            | 5%     |
| Sustainability         | 5%     |
| Total                  | 100%   |

## 16. Impact per Rupee

A signature metric should be **Expected Impact per ₹1 Lakh**. It should be used alongside need, outcomes, feasibility, sustainability and portfolio constraints—not as a standalone decision rule.

| Project | Cost  | Impact | Impact / ₹1L |
|---------|-------|--------|--------------|
| A       | ₹1 Cr | 82     | 0.82         |
| B       | ₹50L  | 78     | 1.56         |
| C       | ₹2 Cr | 95     | 0.48         |

## 17. Portfolio Optimization

Ranking projects individually does not guarantee the best portfolio. A set of highly ranked projects may exceed the budget or over-concentrate funding in one sector or region.

Use Google OR-Tools to maximize expected portfolio impact subject to the available budget and configurable constraints.

$x_i = 1$  if project  $i$  is selected  
 $x_i = 0$  otherwise

Maximize:  
 $\sum \text{ImpactScore}_i \times x_i$

Subject to:  
 $\sum \text{Cost}_i \times x_i \leq \text{AvailableBudget}$

## 18. Geographic Intelligence

The dashboard should contain an India/district-level map showing need, projects and selected allocations. Possible layers include high-need regions, proposed projects, selected projects, sector concentration, rural/urban distribution and estimated unmet need addressed.

## 19. “What Should We Fund Here?”

A powerful interaction is allowing the CSR manager to select a district and immediately see its highest-priority needs and matching proposals.

The system connects **need** → **intervention** → **project** → **expected impact** → **funding**.

## 20. Counterfactual What-If Engine

The What-If Engine should answer: “**What changes if we allocate our money differently?**” rather than merely moving sliders.

Example: move ₹50L from Environment to Water and show estimated changes in beneficiaries, gap addressed, expected overall impact and geographic concentration.

Current Portfolio  
 Healthcare ₹2 Cr  
 Education ₹1.5 Cr  
 Water ₹1 Cr  
 Environment ₹50L

Scenario:  
 Move ₹50L from Environment → Water

Estimated effect:  
 Beneficiaries +3,200  
 Water gap addressed +8%  
 Expected overall impact +5.7%  
 Geographic concentration +2%

## 21. Budget What-If

The CSR manager can change the available budget, such as ■5 Crore → ■3 Crore, rerun optimization and understand which projects leave the portfolio and what expected impact is lost.

## 22. Explainable Recommendations

Every recommendation should provide reasons such as high unmet need, beneficiary reach, cost efficiency, CSR alignment, location, constraint contribution and compliance status.

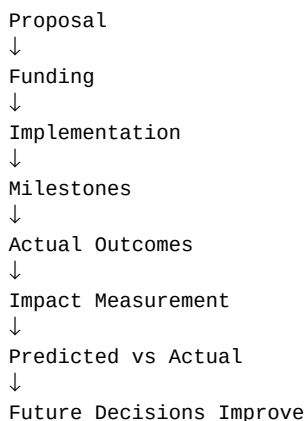
The explanation should be derived from actual scoring and rule data so the decision is traceable.

## 23. Human-in-the-Loop

AI should recommend, not autonomously make irreversible CSR decisions. Reviewers can override classification, modify scoring weights, change constraints, review flags, reject recommendations and approve final allocations.

## 24. Impact Feedback Loop

ImpactSphere can evolve from a one-time allocation tool into a continuous system: Proposal → Funding → Implementation → Milestones → Actual Outcomes → Impact Measurement → Predicted vs Actual → Future Decisions Improve.



## 25. Predicted vs Actual Impact

Record predicted beneficiaries/outcomes against actual results. Over multiple projects, historical outcomes can help calibrate future estimates.

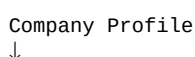
The system must not claim that it guarantees causal social impact; actual impact depends on implementation, context, time horizon and measurement.

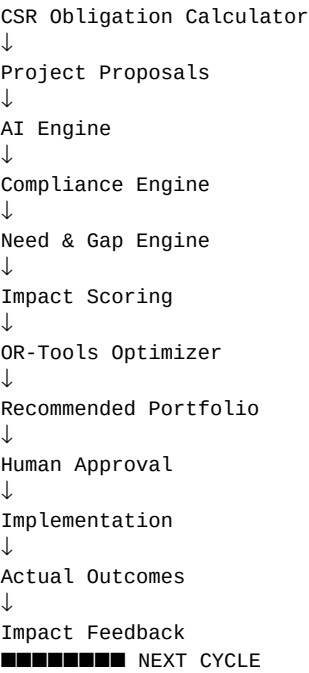
## 26. Community Voice

A future version can incorporate structured community feedback through surveys or post-project assessments, combining government data, project data, actual outcomes and community feedback.

## 27. Complete System Architecture

Overall flow: Company Profile → CSR Obligation Calculator → Project Proposals → AI Engine → Compliance Engine → Need & Gap Engine → Impact Scoring → OR-Tools Optimizer → Recommended Portfolio → Human Approval → Implementation → Actual Outcomes → Impact Feedback.





## 28. Technology Stack

| Layer           | Technology                                           |
|-----------------|------------------------------------------------------|
| Frontend        | Next.js, React, TypeScript                           |
| UI              | Tailwind CSS, shadcn/ui                              |
| Charts          | Recharts                                             |
| Maps            | Leaflet + OpenStreetMap / Mapbox                     |
| Backend         | Python + FastAPI + Pydantic                          |
| AI/ML           | Python, scikit-learn, Sentence Transformers, LLM API |
| PDF             | pypdf and OCR where required                         |
| Optimization    | Google OR-Tools                                      |
| Database        | PostgreSQL + pgvector                                |
| Storage         | Supabase Storage / S3-compatible storage             |
| Version Control | Git + GitHub                                         |

## 29. Four-Person Team Division

| Member     | Ownership             | Responsibilities                                                                                                |
|------------|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| Krishna    | AI / ML               | Dataset, PDF processing, extraction, classification, embeddings, training, evaluation, confidence, AI endpoints |
| Subhashini | CSR Compliance        | Section 135, Schedule VII, calculations, exclusions, agencies, CSR-1, rule DB, compliance API                   |
| Manasa     | Impact + Optimization | Need Index, unmet need, scoring, impact/■■■, OR-Tools, geographic constraints, What-If                          |
| Jwala      | Frontend + Product    | Dashboard, upload UI, maps, portfolio view, What-If UI, explanations, approval workflow                         |



## 30. API Architecture

AI  
POST /ai/extract-proposal  
POST /ai/classify  
POST /ai/analyze

Compliance  
POST /compliance/check  
POST /compliance/calculate-csr  
GET /compliance/rules

Impact  
POST /impact/score  
POST /impact/need-index  
POST /impact/unmet-need

Optimization  
POST /optimization/run  
POST /optimization/what-if

## 31. Proposed Database

users  
companies  
csr\_policies  
csr\_rules  
projects  
proposals  
project\_scores  
budgets  
allocations  
implementing\_agencies  
documents  
milestones  
impact\_metrics  
need\_indicators  
regional\_need\_scores  
audit\_logs

Future:  
project\_outcomes  
community\_feedback  
intervention\_coverage  
optimization\_runs  
what\_if\_scenarios

## 32. Recommended GitHub Structure

```
ImpactSphere/  
■■■ frontend/  
■■■ backend/  
■ ■■■ app/  
■ ■ ■■■ api/  
■ ■ ■■■ ai/  
■ ■ ■■■ compliance/  
■ ■ ■■■ impact/  
■ ■ ■■■ optimization/  
■ ■ ■■■ models/  
■ ■ ■■■ schemas/  
■ ■ ■■■ services/  
■ ■■■ requirements.txt  
■■■ data/  
■ ■■■ raw/  
■ ■■■ processed/  
■ ■■■ train.csv  
■ ■■■ validation.csv  
■ ■■■ test.csv
```

■■■ models/  
■■■ docs/  
■■■ tests/  
■■■ .env.example  
■■■ README.md  
■■■ docker-compose.yml

## 33. Hackathon MVP

Do not attempt the entire long-term vision during the hackathon. Build one complete vertical slice: 5–10 regions, need dataset, 20 CSR proposals, AI extraction/classification, compliance, need score, impact score, OR-Tools optimization, map, What-If simulation and explanation.

## 34. Suggested Judge Demo

Enter ■5 Crore budget → load 20 proposals → show AI extraction → compliance screening → district need → apply constraints → Optimize → explain a selected project → change budget to ■3 Crore → run What-If → inspect a high-need district → show expected beneficiaries, need addressed, impact score and budget utilisation.

## 35. Novelty Positioning

Do not rely on an unsupported claim that no other CSR platform exists. Position the novelty as the integration of **need + gap + proposal intelligence + compliance + impact + optimization + feedback** into a single decision loop.

The central question becomes: “**What is the most socially meaningful use of the next ■1 of CSR money?**”

## 36. Real-Life Implementation Roadmap

**Phase 1 — Hackathon:** synthetic proposals + selected public indicators + end-to-end optimization demo.

**Phase 2 — Pilot:** partner with a company and NGOs for a limited set of districts/projects.

**Phase 3 — Data Integration:** public development indicators, historical CSR projects and outcome data.

**Phase 4 — Impact Feedback:** predicted vs actual outcome tracking and calibration.

**Phase 5 — Enterprise:** multi-year planning, reporting, verification, outcome tracking and advanced optimization.

## 37. Responsible AI

Need scores should be traceable to indicators. A low score must not be described as a judgement about people. Beneficiary count should not be the only proxy for impact. AI classifications should include confidence and evidence. Ambiguous legal cases should go to human review. Predicted impact must be labelled as an estimate.

## 38. Risks and Mitigations

| Risk                            | Mitigation                           |
|---------------------------------|--------------------------------------|
| AI misclassification            | Confidence threshold + human review  |
| Poor proposal quality           | Validation + missing-field warnings  |
| Biased need data                | Transparent indicators + methodology |
| Duplicate interventions         | Coverage/gap analysis                |
| Regulatory changes              | Versioned CSR rule database          |
| Incorrect legal interpretation  | Human/professional review            |
| Prediction differs from reality | Outcome feedback and calibration     |

| Risk                | Mitigation                           |
|---------------------|--------------------------------------|
| Over-reliance on AI | Deterministic rules + human approval |

## 39. Success Metrics

| Area         | Metrics                                                                                |
|--------------|----------------------------------------------------------------------------------------|
| Operational  | Proposal processing time, shortlist time, number of proposals evaluated                |
| AI/ML        | Accuracy, precision, recall, F1-score, low-confidence rate                             |
| Optimization | Budget utilisation, expected impact, impact/■, geographic concentration, need coverage |
| Social       | High-need regions supported, estimated beneficiaries, unmet need addressed             |
| Long term    | Predicted outcome vs actual outcome                                                    |

## 40. Important Limitation: Expected Impact

ImpactSphere should not claim that its AI knows with certainty which project will improve people's lives. It estimates expected contribution using proposal information, need indicators, historical outcomes where available and configurable assumptions.

Actual social impact depends on implementation quality, context, time horizon and measurement.

## 41. Final Product Definition

**ImpactSphere — AI-Powered CSR Need-to-Impact Allocation & Optimization Platform.**

The platform connects community need, unmet gaps, CSR project proposals, AI, CSR compliance, impact scoring, portfolio optimization and outcome feedback.

## 42. Final Vision

ImpactSphere aims to shift CSR from “How much did we spend?” to “What problem did we address?” and ultimately to “Did people's lives actually improve?”.

## 43. Selected Reference Basis

The CSR/legal discussion should be validated against the latest applicable legislation, rules, notifications and official guidance before production use. The platform should maintain source, effective date and version metadata for each rule.

- Companies Act, 2013 — Section 135.
- Companies Act, 2013 — Schedule VII.
- Companies (CSR Policy) Rules, 2014, as amended.
- Ministry of Corporate Affairs — CSR FAQ and official CSR notifications.
- NITI Aayog — Aspirational Districts Programme and associated public development indicators.

**Disclaimer:** ImpactSphere is a decision-support prototype. It is not a substitute for legal, tax, accounting or professional CSR compliance advice. Regulatory rules can change and should be reviewed before real-world deployment.